

SOUTHERN UNIVERSITY
AND AGRICULTURAL & MECHANICAL COLLEGE
DEPARTMENT OF BIOLOGICAL SCIENCES

ECOLOGY 404B: ECOLOGY IN A DIGITAL AGE

Spring 2026 | Fully Asynchronous | "Where Bayous Meet Bytes."

✿ Dr. Mary K. Beals, Ph.D.

🌿 Fisher Hall 224A (225) 771-3012

Email: mary.beals@sus.edu (response time: <48 hrs.)

Eco-Cafés: 2nd Tuesday each month (Zoom link on Padlet).

☕ Office Hours: By appointment (or whenever the coffee's fresh!)

COURSE POLICIES & ESSENTIALS

Course Description

Explore Louisiana's ecological systems through virtual fieldwork, AI tools, and real-world problem solving. This course emphasizes populations, communities, ecosystems, and human-environment interactions—balancing scientific principles with creativity, storytelling, and applied problem-solving.

Required Textbook & Materials

- Molles Jr., M.C. & Simon, A.S. Ecology: Concepts & Applications (9th ed.) – McGraw-Hill Connect eBook.
- Technology Tools: Canvas, Padlet, NotebookLM, Notebook.ai, CoPilot

Course Goals & Objectives

By the end of this course, students will:

- Understand ecological concepts at the individual, population, community, and ecosystem levels.
- Apply ecological principles to real-world environmental challenges.
- Use AI tools, simulations, and digital platforms for ecological inquiry.
- Develop creative science communication skills (memes, podcasts, infographics).
- Engage in informed discussions about ecological and environmental issues.

Grading

Category	%	Examples
Exams	30%	Textbook and notes based
Labs	30%	Eco-detective missions
Creativity	20%	Memes, infographics, haikus
Engagement	20%	Padlet, discussion boards

Grading Scale: A = 90–100 | B = 80–89 | C = 70–79 | D = 60–69 | F < 60

Course Policies & Expectations

Late Work – Students may use 2 Grace Tokens for deadline extensions (redeemable via meme submission). After tokens are used, late work will not be accepted.

AI Usage Policy – AI is welcome as a partner in your learning. You must:

- Acknowledge AI support with a citation (e.g., “ChatGPT drafted this outline; analysis is mine”).
- Demonstrate your own critical thinking in all submissions.
- Avoid presenting AI-generated work as solely your own (violates SU’s Academic Integrity policy).

Dr. Mary's Survival Guide

- ♦ **Emails:**
 - » Subject line: *"Help: My brain is a nutrient-poor wetland!"*
 - » Attach memes for faster replies
- ♦ **AI Use:**
 - » Cite tools like: *"ChatGPT drafted this outline; analysis is mine."*
- ♦ **Grace Tokens:** 2 no-questions-asked extensions (redeem via meme submission).

Important Institutional Policies



Student Code of Conduct & Disruption of the Academic Process

In general, students are expected to conduct themselves in an appropriate manner that is consistent with the SUBR Student Code of Conduct. The University's Student Code of Conduct (Codes 1.6 and 2.2) broadly defines disruption/obstruction as any major/minor action which obstructs, or attempts to obstruct, an official University function, such as teaching, research, administration, or other campus activity.

[Disruption Policy](#) (link to PDF).

Academic Dishonesty

Southern University and A&M College's *Student Code of Conduct* (Code 1.1) identifies academic dishonesty as any deliberate attempt to gain an unfair advantage in academic work. Examples of academic dishonesty include cheating, falsification of information, fraud, plagiarism, and unauthorized access to academic records, providing information, material, or other assistance with knowledge that such assistance could be used in violation of the *Student Code of Conduct* or other University policies, or providing false information in connection with any inquiry regarding academic dishonesty.

[Academic Dishonesty Policy](#) - link to PDF

Academic Grievances

Southern University and A&M College students may seek redress in academic decisions when they believe the decision is unfair or ungrounded. The academic grievance procedure provides prompt and equitable resolutions to student academic grievances. Classroom related matters should be subjected to these procedures only when the grievance cannot be settled in the ordinary course of immediate post-class discussion. Academic grievances are not handled by the Office of the Dean of Students.

[Student Academic Grievance Procedures](#) - link to PDF

Office for Compliance

The Office for Compliance oversees ADA, Disability Services, and Title IX for Southern University and A&M College. Therefore, the Office ensures compliance with the Americans with Disabilities Act of 1990, its Amending Act of 2008,

Section 504 of the Rehabilitation Act of 1973, Title IX of the 1972 Education Amendments, the Pregnant Workers Fairness Act, the Fair Housing Act of 1988, the Jeanne Clery Disclosure of Campus Security Policy and Campus Crime Statistics Act, the Louisiana Campus Accountability and Safety Act, and any other relevant federal or state laws or regulations.

All matters concerning the Louisiana Governmental Code of Ethics are handled by the Southern University System Office for Compliance. Please visit: [Ethics | Southern University System](#).

Office for Compliance

(225) 771-2008

A.C. Blanks Hall, Room 246

<https://www.subr.edu/page/overview-office-compliance>

Disability Services - The Southern University System and its institutions are committed to providing reasonable accommodations for all students with disabilities in compliance with the Americans with Disabilities Act of 1990, its Amending Act of 2008, and Section 504 of the Rehabilitation Act of 1973. In order to request accommodation, students are required to register with the Disability Services Coordinator. It is important to note that accommodation is not retroactive; therefore, students must register and renew each semester as early as possible.

Disability Services Coordinator

Carla Popularas, M.J.

(P) 225-771-3546

(F) 225-771-3949

A.C. Blanks Hall, Room 246

subr.services@sus.edu

<https://www.subr.edu/page/disability-services>

Notice of Non-Discrimination - Title IX is a federal law that prohibits discrimination on the basis of sex in any federally funded education program or activity. Title IX prohibits use of federal money to support sex discrimination in education programs and provides individuals protection against such practices.

In compliance with federal law and USDOE federal guidance, including provisions of Title VII of the Civil Rights Act of 1964 (Title VII), Title IX of the Education Amendments of 1972 (Title IX), Section 503 and 504 of the Rehabilitation Act of

1973, the Americans with Disabilities Act (ADA) of 1990, the ADA Amendments Act of 2008, the Age Discrimination in Employment Act of 1967 (ADEA), the Pregnant Workers Fairness Act (PWFA), Executive Order 14173, the Vietnam Era Veterans Readjustment Assistance Act of 1974 as amended by the Jobs for Veterans Act, the Uniformed Services Employment and Reemployment Rights Act, as amended, and the Genetic Information Nondiscrimination Act of 2008, an institution of the Southern University System shall not discriminate against individuals on the basis of their race, sex, sexual orientation or preference, religion, color, nation or ethnic origin, age, disability, military service, covered veteran's status, or genetic information in its administration of education policies, programs, or activities; admission policies; scholarship and loan programs; athletic or other institution-administered programs; or employment.

As part of their commitment to maintaining a community free of discrimination, and in compliance with Title IX's mandate, institutions of the Southern University System shall address allegations of power-based violence and sexual misconduct, including sexual harassment and sexual assault, in a timely and effective manner. Further, institutions of the Southern University System will provide resources as needed for affected persons (Reporters, Complainants, Respondents, and third parties within the institution's community) and will not tolerate retaliation against any person who reports or participates in the investigation of alleged power-based violence or sex discrimination.

Any member of the Southern University and A&M College community has the right to raise concerns or make a complaint regarding discrimination without fear of retaliation. All inquiries regarding the application of this statement and related Title IX policies may be referred to:

Jared Francis, J.D.

SUBR Title IX Coordinator

(225) 771-4955

A.C. Blanks Hall, Room 245

Email: subrtitleix@sus.edu

Website: <https://www.sus.edu/page/southern-university-notice-of-nondiscrimination>

and/or

Kretrice Chisley
Human Resources Director
(225) 771-2680

Standards for Satisfactory Academic Progress

The Higher Education Act of 1965 as amended and final regulations set by the United States Department of Education (34CFR668.16) require that institutions of higher education establish reasonable standards of Satisfactory Academic Progress (SAP) as a condition of continuing eligibility for federal aid programs. Financial aid recipients are expected to make reasonable progress as a condition of receiving and continuing to receive student financial aid. Student progress is assessed according to both qualitative and quantitative measures. The University has developed this policy to provide a framework for monitoring and determining a student's Satisfactory Academic Progress in accordance with Federal and Institutional requirements. This policy applies to all new, transfer, re-entry, re-admit with transfer work, and continuing students at Southern University.

For additional information, please see the [Satisfactory Academic Progress policy](#).

Student Email Policy

Email is a universal service that has greatly enhanced communication both internally within the Southern University and A&M College community and externally to users, including prospective students, alumni, and the public at large. The purpose of the University's general email policy is to describe the appropriate use of University email facilities, associated responsibilities, and rights of all users of University email facilities and official Southern University and A&M College email accounts. This student email policy is a supplement to the University's general email policy. [Student Email Policy](#) - link to PDF

Tobacco Free Campus Policy

The Southern University System (SUS) maintains a 100% Tobacco-Free Policy. Smoking and tobacco use of any kind will be prohibited on all SUS campuses and/or other properties owned and/or leased locations/premises; all internal and external areas, parking garages, and parking lots; all entrances and exits; and in all SUS owned and/or leased vehicles. Students may not smoke in their own or others'

vehicles when the vehicles are parked on SUS properties. This policy will be in place at all Southern University and A&M College sponsored events-both on our premises and at external locations hosting such events, including non-University hours and will further apply to all faculty, staff, students and visitors. [SUBR Tobacco Free Policy](#) - [link to PDF](#)

Civility in Academic Settings

Civil behavior enhances the academic setting and is expected at all times. Respect and courtesy create a pleasant environment and improve learning for all. Use of profanity or cuss words in the classroom will not be tolerated and violators of this language courtesy policy will be punished by disciplinary actions. It is the policy of Southern University to administer its academic and employment programs and related supporting services in a manner which does not discriminate on the basis of gender, race, color, national origin, age, religion, disability, or marital status. Appropriate action will be taken in the event that a student violates the above policies.

Student Learning & Licensure by Watermark (replaced LiveText)

The assessment of learning outcomes in the academic programs is comprehensive and inclusive of all academic programs. The faculty within each degree program identify student learning outcomes for all courses offered each semester, as it relates to the program. Rubrics are then developed for all courses taught within each degree curriculum. Faculty members who are teaching these classes make assignments for each course per semester and assess student learning outcome. In each educational program, faculty members, with the leadership of the chair, convene annually to determine the extent to which student performance matched expectations towards program learning outcomes. Remedial measures are taken to enhance instruction and student learning as needed thereby closing the loop for each cycle.

Student Learning and Licensure is a modern and flexible platform for assessing student learning in the classroom, field, and co-curricular activities, report on learning outcomes achievement, and provide students with timelines of work and ePortfolios to track and share their learning journeys.

For SLL assistance, please contact:

Chrisena Williams-Brown, MBA/TM
Associate Director of Planning and Assessment
Office of Strategic Planning, Policy, and Institutional Effectiveness
Room 208 Augustus Blanks Hall
Email: chrisena.williams@sus.edu
Office Phone: 225-771-2274

Watermark-Course Evaluation

Course Evaluations & Surveys by Watermark is an easy-to-use course evaluation system that integrates directly with Canvas. You will receive an email when the evaluation survey is available. The email will provide more information as well as a link and instructions on how to access the evaluation. We encourage you to be open and honest in these evaluations. Your identity will be kept private, and your instructor will not see the results of your evaluation until classes have ended. No identifying information is ever directly connected to the feedback you submit.

Follett Access Textbook Program

Follett Access is a partnership between the Southern University System and the SUBR Bookstore that allows you to rent textbooks at a reduced cost. The charge is posted directly to your SUBR tuition and fees bill and are paid along with your University charges. Students are able to opt out of Follett Access. If you choose to opt out of the program, be aware that you are opting out for the entire semester schedule. You may not opt out for individual courses. Fees will be refunded on your SUBR account after the add/drop date for each semester. To opt out of Follett Access, create an account on the [Follett Customer Portal](#) and follow prompts to opt out.

For more information about the program, including information about accessing your textbooks, opt-out deadlines, and more resources, refer to the [Follett Access Textbook Program guide](#). For assistance, contact the Bookstore at (225) 771-1472 or 0622txtmgr@follett.com.

Mental Health Resources

Mental health resources and support are available from the [University Counseling Center \(UCC\)](#). The staff at UCC is trained to help you navigate the challenges and opportunities that arise for college students. The UCC provides free individual/couple counseling, crisis intervention, and other psychological services in a safe and confidential setting.

For mental health resources or to schedule an appointment, students may contact the University Counseling Center at counselingcenter@subr.edu or by calling (225) 771-2480. If you are experiencing an emergency, call 911.

ATTENDANCE POLICY (*Effective Fall 2024*)

Class attendance is mandatory for all enrolled students. Each student must know an instructor's attendance policy as stated in or attached to the course syllabus and adhere thereto. Students must attend courses in the delivery mode presented in the course description (face-to-face, synchronous online, or asynchronous online). Students who miss any classes, regardless of the reason, shall be responsible for making up examinations, obtaining lecture notes, and completing any other required work that was missed.

Unexcused Absences

- Students will be permitted unexcused absences based on the per credit hour attributable to a given course, as noted below.
 - ◆ Four (4) credit hour course: six (6) unexcused absences
 - ◆ Three (3) credit hour course: five (5) unexcused absences
 - ◆ Two (2) credit hour course: 3 (three) unexcused absences
 - ◆ One (1) credit hour course: 2 (two) unexcused absences
- When a student exceeds the permitted number of unexcused absences, the instructor may lower the student's final grade by five (5) points on a 100-point scale.
- **Makeup work may be facilitated at the instructor's discretion. (As such, this instructor WILL NOT ALLOW MAKEUP WORK FOR UNEXCUSED ABSENCES.)**

Excused Absences

The course instructor will determine the validity of a student's reason(s) for absences and will excuse (and assist with) makeup work for those students whose

absences are excusable. Instructors shall not unreasonably deny, delay, or withhold approval of a student's request for allowable absence excusal.

- **Request for Excuse**

- ◆ Students must inform their instructors prior to an absence, to the extent possible, or within three business days after the student returns to class. Whether to excuse an absence for which such excuse is requested later than three business days after a student's return to class is subject to the instructor's discretion.
- ◆ Notice of student absences should be given in person when possible. If notice is provided via email, the student should follow up with the instructor in person if the instructor does not respond.
- ◆ Approval requests for absences due to participation in university-sponsored activities should be initiated in advance by the sponsoring department or organization and approved by the applicable College Dean or the Office of Academic Affairs.

- **Verification**

- ◆ Official documentation that is both appropriate and verifiable must be provided by the student for an absence to be excused.
- ◆ In the event a student has been hospitalized, is injured or ill, or has experienced mental health issues, documentation is required from the treating physician.
- ◆ Students must inform their instructors prior to the absence or within three business days after their return to class. Requests for absence excusal after this period will be honored at the instructor's discretion.
- ◆ Notice of student absences should be given in person when possible.

- **Makeup Work**

- ◆ As applicable, an instructor will facilitate makeup work for major assignments and assessments, such as essays and tests, within three business days after a student's return.
- ◆ Makeup work for minor assignments, such as quizzes and in-class participation assignments may be facilitated at the instructor's discretion.
- ◆ Should an instructor decide not to allow makeup work for minor assignments, such minor assignments shall be deemed waived and non-completion shall not adversely impact a student's grade.

- ◆ An instructor shall assign a deadline for the completion of makeup work for either major or minor assignments, which deadline shall be reasonable, commensurate with the nature of the applicable assignment(s), and accounts for any accommodation required by law or regulation. Subject to the foregoing, valid reasons for absences may include, but are not limited to, the following:
 - ◆ Serious or infectious illnesses, injury, or other health reasons;
 - ◆ Reasons relative to the Office of Disability Services and related accommodations;
 - ◆ Religious observances;
 - ◆ Serious emergencies involving members of the student's household or immediate family;
 - ◆ Court-imposed legal obligations such as subpoenas or jury duty;
 - ◆ Voting in local, state, or federal elections;
 - ◆ Military obligations, including, but not limited to, deployment, reserve service, and to include official Reserve Officer Training Corps (ROTC) events;
 - ◆ Hazardous weather conditions or effects;
 - ◆ Interviews at career fairs (SUBR sponsored), graduate schools, or professional schools;
 - ◆ Varsity athletic, band, and other university recognized activities in which the student is representing the university; and
 - ◆ Circumstances that are determined to be beyond the student's control.

Appealing an Unexcused Absence

To contest an unexcused absence, the student must appeal to the Office of the Dean who has oversight for the respective course. The Dean will consider the case and attempt to resolve the problem. If the case cannot be satisfactorily resolved at that level, the student may utilize the established academic grievance process

SUS Board Artificial Intelligence (AI) Usage Policies (adopted Fall 2025)

<https://www.sus.edu/assets/sus/systempolicies/System-Policies/01/1-004.pdf>
(Comprehensive Use Policy)

<https://www.sus.edu/assets/sus/systempolicies/System-Policies/08/8-010.pdf>
(Umbrella Policy)

Summary

The use of Artificial Intelligence (AI) tools, such as ChatGPT, Grammarly, or other similar technologies, is permitted under the following guidelines to ensure academic integrity and the development of your skills:

a. Appropriate Use of AI

- i. AI tools may be used as a supplementary resource to enhance your learning, such as for grammar and spell-checking, generating ideas, or assisting with formatting.
- ii. AI tools may not be used to generate content that you submit as your own work unless explicitly authorized by the instructor.

b. Transparency and Disclosure

- i. Any use of AI in completing an assignment must be disclosed.
- ii. Students are required to include a detailed statement in their submission specifying:
 1. The type of AI tool used.
 2. The exact prompt or input given to the AI tool.
 3. The specific output or response generated by the AI tool that was incorporated into the assignment.
- iii. Example:
 1. Tool: ChatGPT
 2. Prompt: "Provide a summary of the causes of World War I."
 3. Output: "World War I was caused by a combination of militarism, alliances, imperialism, and nationalism. These factors, along with the assassination of Archduke Franz Ferdinand, led to a global conflict."

c. Prohibited Use of AI

- i. AI tools must not be used to complete assessments, quizzes, exams, or any other work where independent effort is required.
- ii. Misrepresentation of AI-generated content as original work will be considered academic dishonesty and subject to disciplinary action under the university's academic integrity policy.

d. Responsibility for Submitted Work

- i. You are fully responsible for the accuracy, originality, and quality of the work you submit, regardless of AI assistance. Always review and revise content generated by AI to ensure it aligns with course requirements and your understanding of the subject matter.

e. Instructor-Specific Guidance

- i. Individual instructors may provide specific guidelines for or restrictions on the use of AI tools in their courses. It is your responsibility to adhere to these guidelines as stated in the course syllabus or other communications.

By following this policy, students can responsibly incorporate AI tools into their learning experience while maintaining the integrity and academic standards of the course. If you have any questions about this policy or the appropriate use of AI, please contact your instructor.

NOTICE: *The instructor reserves the right to make changes in the syllabus, course content or instructional techniques in the event it becomes necessary. Written notice of any changes will be given through SUS e-mail and/or Canvas announcements.*



COURSE SCHEDULE

Module	Dates	Theme & Topics	Key Activities / Assessments	Notes
1. Mapping Our Roots	Jan 12–23	Natural History, Evolution, Louisiana Ecosystems	Padlet Ecosystem Map, NoteBookLM Reading Annotation	No Class Jan 19 (MLK)
2. Coding Resilience	Jan 26–Feb 6	Temperature & Water Stress, Climate Adaptation	AI Debate, Digital Water Use Tracker	
3. Data-Driven Populations	Feb 9–20	Population Growth, Life Histories	Simulation Lab, Perusall Review	No Class Feb 17–18 (Mardi Gras)
4. Networks & Niches	Feb 23–Mar 6	Competition, Mutualism, Community Ecology	Invasive Species Roleplay, Pollinator Media	
—Spring Break—	Mar 9–13	No Classes		University Break
5. Systems in Motion	Mar 16–27	Nutrient Cycles, Succession	Nitrogen Escape Room, Succession Map	
6. Climate in the Cloud	Mar 30–Apr 17	Climate Change, Landscape Ecology	Baton Rouge 2050 Project, Podcast Review	No Class Apr 3 (Good Friday)
7. Digital Bayou Showcase	Apr 20–May 6	Synthesis, Applied Ecology, Capstone	Final Project Presentations	Seniors: Apr 27–May 3
Final Exams	May 7–12	Cumulative / Project Defense	Final Submission	